Velocity graphs and components

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Read a velocity graph and describe motion in two directions.

SUCCESS CRITERIA - I can...

1. Use signed velocity-time area to find displacement.
2. Represent horizontal and vertical components separately.
3. Distinguish graph height, slope and vector magnitude.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “component” and “axis” to explain your answer. Sentence starters are optional.

component: the part of a vector along one chosen axis

axis: a reference line used to measure a direction

displacement: change from initial position to final position, with direction

01 Warm-up: bring back an earlier idea

Positions are **18**, **12**, **6 m** at **0**, **2**, **4 s**.
1. **Predict position** at **8 s**.
2. **Give the rate**, something you are treating as true and a future test.

Sentence starter: I predict … . I used … to make this prediction because … .

Marzano check - circle: ① ② ③

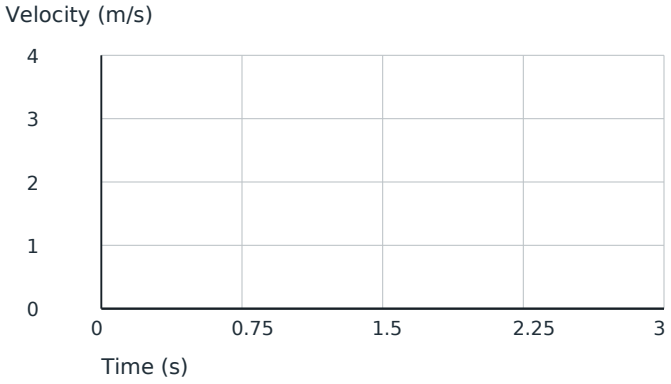
02 Worked example: explain the prediction

Reference: $v_x = v \cos\theta$; $v_y = v \sin\theta$.
θ measured from +x; degree mode. Signed velocity-time area gives displacement; total absolute area gives distance.

Worked reasoning: A cart travels at **3 m/s** *east* for **2 s**. The rectangle under its velocity-time graph has area **3×2=6 m**, its displacement.

Your turn: Draw the cart's constant velocity on the axes. **Shade the area** between the line and the time axis. **Explain why multiplying the two** axis units gives meters.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



Marzano check - circle: ① ② ③

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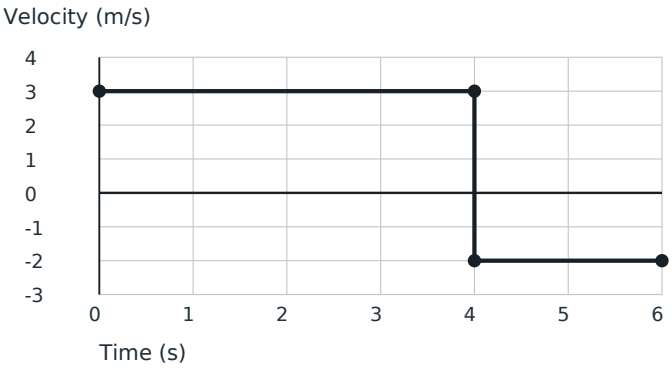
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Add signed areas

1. **Use the graph**.
2. **Find displacement** and distance from **0** to **6 s**.
3. **Describe the motion** on each interval.

Sentence starter: I found ... [unit]. This value means ... in this situation.

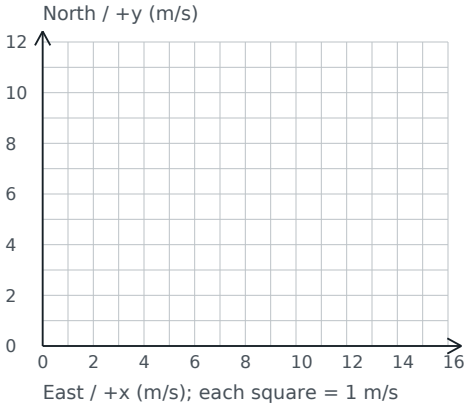


Marzano check - circle: ① ② ③

04 Resolve a northeast velocity

A drone flies at **14 m/s**, **35°** *north* of *east*.
1. **Sketch its velocity** and component triangle on the grid.
2. **Predict which component** is larger, then **calculate both**.

Sentence starter: Both ... . They differ in ... because ... .

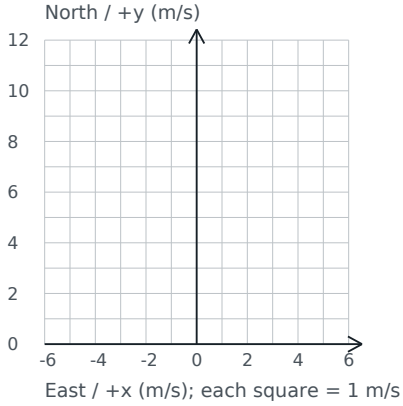


Marzano check - circle: ① ② ③

05 Change the reference axis

A boat travels at **10 m/s**, **20°** *west* of *north*. *East* is +x and *north* is +y.
1. **Sketch the velocity**, mark the angle from *north*, and **find both signed components**.
2. **Explain your sine**/cosine choices.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .



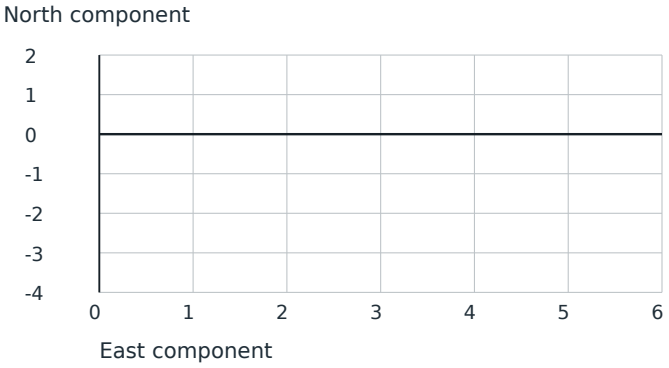
Marzano check - circle: ① ② ③

06 Draw velocity and displacement

An object has v=(**+4**,**-3**) m/s for **2 s**. **Draw the velocity vector** and **calculate its speed**, displacement vector, and displacement magnitude.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Two horizontal graphs

1. **Explain the difference** between a horizontal line at **+5** on a position-time graph and a horizontal line at **+5** on a velocity-time graph.
2. Include units.

Sentence starter: Both ... . They differ in ... because ... .

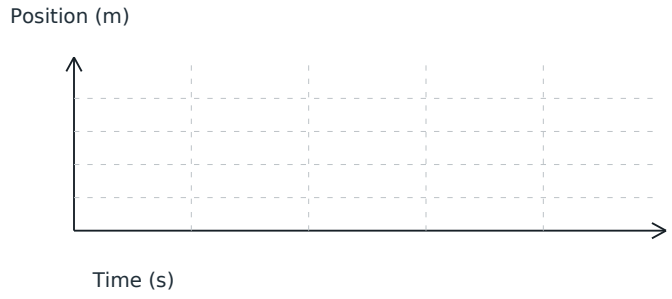
Marzano check - circle: 1 2 3

08 Review: connect an earlier idea

A straight position-time segment goes from (**1 s**, **-3 m**) to (**5 s**, **17 m**). **Find velocity**, interpret its units and **predict x** at **7 s** if motion continues.
Label the diagram with the given positions and any prediction.

Word help: velocity = rate of position change, with direction

Sentence starter: I predict ... . I used ... to make this prediction because ... .



Marzano check - circle: 1 2 3

09 Exit check

A velocity graph stays at **+2 m/s** for **3 s**, then **-1 m/s** for **2 s**.
1. **Find displacement** and distance.
2. **State the information needed** to interpret a vector angle of **30°**.

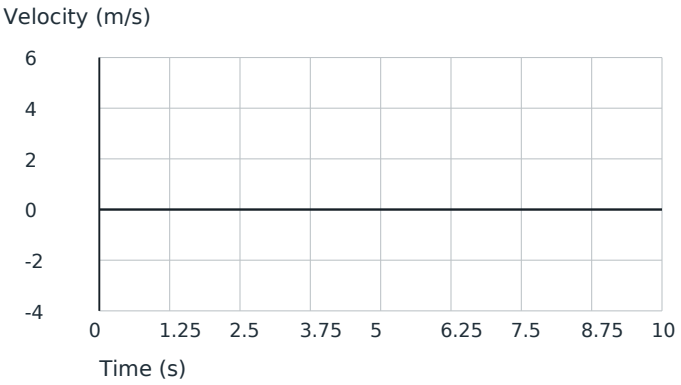
Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

10 STRETCH: test the prediction

A cart moves at **+4 m/s** for **3 s**, then **-2 m/s**. **How long** must the second interval last to return to its start? **Sketch the graph** and **find average speed** for the full trip.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3